



Product Catalogue

Compounds

Epoxy resins

Hardeners

01 / 03

Compounds

Systems, processing and technical properties

Lapox				Processing techniques										Physical/Chemical properties			
Systems	Type of system	Nature	Mixing ratio	CVC	APG	Vacuum potting	Filament winding	Pultrusion	Hand lay-up	Impregnation/VPI	Encapsulation	Prepregging	Bonding	Mix viscosity	Pot life	Gel time	Density of c omponents
	Unf/Prf	Hot cure/Cold cure/ Outdoor/Indoor	R/CA/Acc/Fix/Flr/Sol														Resin Curing agent
			pbw											mPa s/°C	hr/°C	min/°C	g/cm³
ARC-15 AH-122	Prf	Hot cure/Indoor	R: 100 CA: 100	●	●									40,000/25 10,500/40 2,000/60		60/100 20/120	1.70 - 1.90 1.70 - 1.90
ARC-16 AH-118 ADP-11 Silica	Unf	Hot cure/Indoor	R: 100 CA: 80 Fix: 0 - 8 Flr: 320 - 350	●	●									5,500/40 3,000/60 1,500/80	24 - 48/25	228/80 12/120	1.13 - 1.18 1.15 - 1.20
ARCH-11 K-3 K-13 Silica	Unf	Hot cure/Outdoor	R: 100 CA: 90 Acc: 0.5 - 2.0 Flr: 300	●										10,000 - 15,000/40 3,000 - 5,000/60 1,000 - 1,500/80	3/60 1/80	4/140 3/150 2/160	1.20 - 1.25 1.36 - 1.40
ARCH-11 AH-214 K-13 Silica	Unf	Hot cure/Outdoor	R: 100 CA: 90 Acc: 0.5 - 2.0 Flr: 270 - 300	●	●									1,000/80	1/80	12 - 13/100	1.20 - 1.25 1.15 - 1.20
ARCH-11 AH-215 Silica	Unf	Hot cure/Outdoor	R: 100 CA: 90 Flr: 270	●	●									1,000/80	1/80	12 - 13/100	1.20 - 1.25 1.15 - 1.20
ARCH-12 AH-214 K-13 Silica	Unf	Hot cure/Outdoor	R: 100 CA: 80 Acc: 0.5 - 2.0 Flr: 270	●	●									6400/25 3000/40 700/60	8/40 4/60	122/80 11/120 5/140	1.14 - 1.20 1.15 - 1.20
ARCH-12 AH-215 Silica	Unf	Hot cure/Outdoor	R: 100 CA: 80 Flr: 270	●	●									6,400/25 3,000/40 700/60	8/40 4/60	122/80 11/120 5/140	1.14 - 1.20 1.15 - 1.20
ARCH-13 AH-215 Silica	Unf	Hot cure/Outdoor	R: 100 CA: 70 Flr: 270	●	●									1,600 - 2,200/80	8/40	90 - 100/100	1.14 - 1.20 1.15 - 1.20
C-17 K-12 K-13 K-14 Silica	Unf	Hot cure/Indoor	R: 100 CA: 100 Acc: 0.5 - 1.5 Fix: 0 - 20 Flr: 400	●	●									41,500/40 1,500/80	14/40 1.5/80	90/80 35/100 12/120	1.15 - 1.20 1.15 - 1.22
C-50 K-61 Silica	Unf	Hot cure/Indoor	R: 100 CA: 80 Flr: 270	●	●									26,000/40 6,800/60 1,000/80	15/40 2/80	285/80 14/120 5/140	1.10 - 1.20 1.15 - 1.25
C-50 K-225 Silica	Unf	Hot cure/Indoor	R: 100 CA: 80 Flr: 270	●	●									50,000/40 11,800/60 2,300/80	48/25	265/80 14/120 4/140	1.10 - 1.20 1.15 - 1.25
C-51 K-6 Silica	Unf	Cold cure/Indoor	R: 100 CA: 10 Flr: 120			●				●				8,500/25	0.5/25	-	1.13 - 1.16 0.97 - 0.99
ARC-18 AH-123	Prf	Hot cure/Indoor	R: 100 CA: 25	●		●								2,500 - 3,500/25	-	30 - 40/100	1.74 - 1.84 1.18 - 1.22
ARC-30 AH-714	Prf	Cold cure/Indoor	R: 100 CA: 25	●		●								1,800/27	2.5/27	90/130 50/140	1.42 - 1.62 0.95 - 0.98
C-2 K-1 Silica	Unf	Hot cure/Indoor	R: 100 CA: 30 Flr: 200	●										4,000/120 2,500/140	1/120 0.5/140	100/130 60/140	1.20 - 1.25 1.50 - 1.55
C-59 K-1 Silica	Unf	Hot cure/Indoor	R: 100 CA: 30 Flr: 200	●										1,900/120 790/130 500/140	6/120	330/130 150/150	1.20 - 1.25 1.50 - 1.55

Processing guideline	Mechanical properties		Thermal properties				Electrical properties			Applications
Cure schedule	Tensile strength Elongation	Flaxural strength Surface strain	Glass transition temperature (Tg)	Thermal conductivity	Thermal class	Coefficient of linear thermal expansion	Breakdown strength (BDV)	Dielectric constant / Relative permittivity	Dielectric dissipation factor (Tan D)	SG/Ins/Bush/IT/DDT/RM/Ecomp/Adh/Paper coating/Eass
	ISO 527-2	ISO 178	DIN 11357-2	ISO 8894-1	IEC 60085	ISO 11359-2	IEC 60243-1	DIN 53483	IEC 60250	
	hr/°C	MPa/%	°C	W/mK	-	(°C) ⁻¹	kV/mm	-	%	
10/80 + 1/140	75 - 85 0.8 - 1.2	120 - 130 -	110 - 120	0.65 - 0.75	H	27 - 30	18 - 22	4.2	< 2	SG/Ins/Bush
10/140	75 - 85 1.1 - 1.3	135 - 155 1.5 - 1.8	105 - 115	0.85 - 0.95	F	31 - 36	18 - 22	4.1	2 - 3	SG/Ins/Bush/IT/DDT
6/80 + 10/140	90 - 100 1.7 - 1.9	150 - 165 2.0 - 2.5	100 - 110	0.80 - 0.90	F	39 - 41	19 - 21	3.8 - 4.2	1.5 - 2.0	SG/Ins/Bush
10/140	90 - 100 1.7 - 1.9	150 - 165 2.0 - 2.5	100 - 110	0.80 - 0.90	F	-	19 - 21	3.8 - 4.2	2 - 3	SG/Ins/Bush/IT
10/140	90 - 100 1.7 - 1.9	150 - 165 2.0 - 2.5	100 - 110	0.80 - 0.90	F	-	19 - 21	3.8 - 4.2	2	SG/Ins/Bush/IT
6/80 + 10/140	70 - 90 1.0 - 1.5	150 - 170 1.7 - 2.0	75 - 85	-	F	-	18 - 22	-	2 - 3	SG/Ins/Bush
6/80 + 10/140	70 - 90 1.0 - 1.5	150 - 170 1.7 - 2.0	75 - 85	-	F	-	18 - 22	-	2 - 3	SG/Ins/Bush
6/80 + 10/140	60 - 70 0.75 - 1.25	-	100 - 110	-	F	-	19 - 21	-	2 - 3	SG/Ins/Bush
8/80 + 8/130	75 - 85 0.9 - 1.1	125 - 135 1.1 - 1.5	85 - 95	0.80 - 0.90	F	31 - 36	18 - 22	4	1.3	Ins/Bush/SG/IT/DDT
6/80 + 8/140	70 - 80 1.0 - 1.3	110 - 125 1.2 - 1.7	105 - 125	0.80 - 0.90	H	35 - 37	18 - 20	4.3	1.7	SG/Ins/Bush
6/80 + 10/130	70 - 80 1.0 - 1.5	110 - 120 1.2 - 1.7	90 - 105	0.80 - 0.90	H	36 - 40	18 - 20	3.8	1.7	SG/Ins/Bush
6/80	35 - 50 -	50 - 60 -	60 - 70	0.54	B	60 - 65	18 - 22	4.4	2 - 3	Adh/Eass
4/80 + 2/130	60 - 70 -	110 - 120 1.5 - 2.5	122 - 132	0.6	F	34 - 36	25	3 - 4	1.3	SG/Ins/Bush
4 - 6/80 - 100	15 - 25 1.5 - 2.5	60 - 90 1.5 - 2.5	60 - 70	-	B	-	18 - 20	-	2 - 3	IT/Eass
12/140	80 - 100 1.0 - 1.4	130 - 150 2.1 - 2.3	115 - 130	-	F	-	22 - 25	-	2 - 3	Ins/Bush/IT
24/140	85 - 95 1.0 - 1.4	135 - 145 2.1 - 2.3	115 - 130	0.75 - 0.90	F	35 - 38	-	4.1	1.5	Ins/Bush/SG/IT/DDT

Lapox				Processing techniques										Physical/Chemical properties			
Systems	Type of system	Nature	Mixing ratio	CVC	APG	Vacuum potting	Filament winding	Pultrusion	Hand lay-up	Impregnation/VPI	Encapsulation	Prepregging	Bonding	Mix viscosity	Pot life	Gel time	Density of c omponents
	Unf/Prf	Hot cure/Cold cure/ Outdoor/Indoor	R/CA/Acc/Fix/Flr/Sol														Resin Curing agent
			pbw											mPa s/°C	hr/°C	min/°C	g/cm³
ARL-111 AH-358 ARS-101	Unf	Hot cure/Indoor	R: 100 CA: 15	●								●		1,300/25	> 8/25	45/120 12/140	0.90 - 1.00 1.02 - 1.06
C-1 K-1 Silica	Unf	Hot cure/Indoor	R: 100 CA: 30 Flr: 200	●										4,000/120 2,500/140	1/120	90/130 50/140	1.20 - 1.24 1.50 - 1.55
ARF-11 K-918 K-13	Unf	Hot cure	R: 100 CA: 85 Acc: 1 - 3				●	●						300 - 500/25	> 8/25	8 - 11/120	1.17 - 1.18 1.18 - 1.22
L-12 AH-113 K-13	Unf	Hot cure	R: 100 CA: 95 Acc: 0.1 - 2.0				●	●						1,900 - 2,100/25	50 - 55/25	-	1.10 - 1.20 1.36 - 1.40
L-12 K-918 K-13	Unf	Hot cure	R: 100 CA: 85 Acc: 0.1 - 2.0				●	●	●					600 - 900/25	80 - 95/25	10 - 12/120	1.10 - 1.20 1.18 - 1.22
L-247 K-918 K-13	Unf	Hot cure	R: 100 CA: 65 Acc: 1 - 3				●	●						-	> 8/25	8 - 10/120	- 1.18 - 1.22
ARC-31 AH-114 AC-13	Unf	Hot cure/Indoor	R: 100 CA: 100 Acc: 0.0 - 0.2							●	●			700 - 750/23	1/80	17/120	1.15 - 1.20 1.15 - 1.22
ARC-35 AH-676	Prf	Hot cure/Indoor	R: 100 CA: 26								●			10,000 - 13,000/25	> 8/25	30/120	1.72 - 1.76 1.00 - 1.05
ARPN-36 K-10 K-86 Solvent	Unf	Hot cure	R: 100 CA: 40 Acc: 1 Sol: 125									●		-	> 8/25	7 - 12/120	1.20 - 1.22 1.30
ARPN-36 K-86 Solvent	Unf	Hot cure	R: 100 CA: 3-6 Sol: 80									●		-	> 8/25	8 -12/120	1.20 - 1.22 1.42
G-11 L-12 K-10 K-86 Solvent	Unf	Hot cure	R: 100 CA: 35 Acc: 1 - 3 Sol: 100 - 150									●		-	6 - 8 weeks/25	-	1.10 - 1.20 1.30
G-10 L-67 K-66 K-13 Solvent	Unf	Hot cure	R: 100 CA: 23 Acc: 0.2 Sol: 10									●		-	3 - 4 weeks/20	8.5/150 3.5/172	- 1.06 - 1.09
FR-4 L-68 K-66 K-13 Solvent	Unf	Hot cure	R: 100 CA: 32 Acc: 1 - 3									●		-	3 - 4 weeks/20	8.5/150 3 - 4/172	0.98 - 1.05 1.06 - 1.09
A-31 AH-717	Unf	Cold cure	R: 100 CA: 80										●	25,000 - 40,000	130 - 150/25	-	1.06 - 1.18 0.92 - 0.98
A-35 K-35	Unf	Hot cure	R: 100 CA: 30										●	400 - 600	3 - 4 weeks/25	-	-
A-38 K-99	Unf	Cold cure	R: 100 CA: 40										●	60,000 - 90,000	35 - 45/25	-	1.65 - 1.75 1.65 - 1.75
C-51 K-6	Unf	Cold cure	R: 100 CA: 10						●				●	-	-	-	1.13 - 1.16 0.97 - 0.99

Processing guideline	Mechanical properties		Thermal properties				Electrical properties			Applications
Cure schedule	Tensile strength Elongation	Flexural strength Surface strain	Glass transition temperature (Tg)	Thermal conductivity	Thermal class	Coefficient of linear thermal expansion	Breakdown strength (BDV)	Dielectric constant / Relative permittivity	Dielectric dissipation factor (Tan D)	SG/Ins/Bush/IT/DDT/RM/Ecomp/Adh/Paper coating/Eass
	ISO 527-2	ISO 178	DIN 11357-2	ISO 8894-1	IEC 60085	ISO 11359-2	IEC 60243-1	DIN 53483	IEC 60250	
hr/°C	MPa/%	MPa/%	°C	W/mK	-	(°C) ⁻¹	kV/mm	-	%	
6/140	-	-	80 - 90	-	F	-	-	-	2.5 - 3.0	Paper coating
16/140	70 - 90 1.0 - 1.4	70 - 100 2.1 - 2.3	115 - 130	-	F	-	22 - 24	-	2 - 3	IT/SG/Bush
4/100 + 4/120	75 - 85 4.0 - 5.0	120 - 140 6.0 - 10.0	115 - 130	-	-	-	-	-	-	Ecomp
2/100 + 2/120 + 8/180	45 - 60 1.5 - 3.0	100 - 140 4.0 - 7.0	165 - 185	-	-	-	18 - 20	3 - 4	2 - 3	Ecomp
2/100 + 2/140	75 -85 4.0 - 5.0	120 - 140 6.0 - 10.0	130 - 140	-	-	-	20 - 25	3.3	0.3	Ecomp
3/80 + 3/120 + 4/160	60 - 80 3.0 - 6.0	120 - 140 5.0 - 10.0	110 - 120	-	-	-	-	-	-	Ecomp
6/80 + 3/120	75 - 85 0.9 - 1.1	125 - 135 1.1 - 1.5	90 - 100	-	F	31 - 36	18 - 22	-	1.3	DDT/Eass/RM
2/140	-	-	70 - 80	-	F	-	-	-	2 - 3	Eass
1 - 2/80 or 15 min/130 (drying) 5/200 (in oven)	-	135 7.0	-	-	-	-	-	-	-	Ecomp
1 - 2/80 or 15 min/130 (drying) 4/120 + 16/160 (in oven)	-	-	-	-	-	-	-	-	-	Ecomp
10 - 15 min/130 or 6 - 10 min/135 (drying) 5/200 (curing)	-	400 - 500 -	150 - 160	-	-	-	15 - 20	4.7 - 5.2	2.0 - 2.5	Ecomp
7 - 9 min/150 or 5 - 6 min/160 or 3 - 5 min/170	≥ 315 (LW)* ≥ 266 (CW)* 2.0 - 3.0	≥ 525 (LW)* ≥ 455 (CW)* 2.0 - 6.0	130 - 140	-	-	-	12 - 14	4.5 - 5.2	2.5 - 3.0	Ecomp
7 - 9 min/150 or 5 - 6 min/160 or 3 - 5 min/170	≥ 315 (LW)* ≥ 266 (CW)* 2.0 - 3.0	≥ 525 (LW)* ≥ 455 (CW)* 2.0 - 6.0	130 - 140	-	-	-	12 - 14	4.5 - 5.2	2.5 - 3.0	Ecomp
4 - 6/40 or 45 - 60 min/65 or 15 min/100	-	-	-	-	-	50 - 60	-	-	-	Adh
24/120 5/140 2/160 1/180	-	-	-	-	-	60	-	-	-	Adh
24 - 28/25 or 1/60	42 - 44 -	-	-	-	-	65	-	-	-	Adh
24/25 or 6/80	55 - 80 -	-	60 - 70	0.19 - 0.20	-	90 - 95	18 - 22	4	2 - 3	Adh

02 / 03

Epoxy resins

Base and specialty epoxy resins

BISPHENOL-F AND BISPHENOL-A/F BLENDS

Bisphenol-F based resins are known for low viscosity, better chemical resistance and a low crystallisation tendency in cold conditions. Atul offers Bisphenol-F based pure and Bisphenol-A/F blend resins in varying viscosities for several applications.

Lapox® grade	Active substance	CAS No.	Colour / APHA	EEW g/eq	Viscosity @ 25°C mPa·s	Recommendations
ARF-11 (XR-40)	Formaldehyde, polymer with (chloromethyl)oxirane and phenol	9003-36-5	Max 200	159 - 175	2,000 - 5,000	A standard Bis-F based epoxy resin recommended for coating applications, composites, construction and electrical casting. CAS # 9003-36-5 42423-25-6
	Phenol, 4,4'-methylenebis-, polymer with (chloromethyl)oxirane	42423-25-6				
ARF-12	Formaldehyde, polymer with (chloromethyl)oxirane and phenol	9003-36-5	Max 200	164 - 172	2,000 - 3,300	A low viscosity Bis-F based epoxy resin recommended for coating applications, composites, construction and electrical casting. CAS # 9003-36-5 42423-25-6
	Phenol, 4,4'-methylenebis-, polymer with (chloromethyl)oxirane	42423-25-6				
ARF-13	Formaldehyde, polymer with (chloromethyl)oxirane and phenol	9003-36-5	Max 200	164 - 172	3,300 - 4,100	A medium viscosity Bis-F based epoxy resin recommended for coating applications, composites, construction and electrical casting. CAS # 9003-36-5 42423-25-6
	Phenol, 4,4'-methylenebis-, polymer with (chloromethyl)oxirane	42423-25-6				
ARF-14	Formaldehyde, polymer with (chloromethyl)oxirane and phenol	9003-36-5	Max 200	159 - 172	5,000 - 7,000	A high viscosity Bis-F based epoxy resin recommended for coating applications, composites, construction and electrical casting. CAS # 9003-36-5 42423-25-6
	Phenol, 4,4'-methylenebis-, polymer with (chloromethyl)oxirane	42423-25-6				
ARF-15	Formaldehyde, polymer with (chloromethyl)oxirane and phenol	9003-36-5	Max 100	156 - 167	1,200 - 1,600	A distilled and pure Bis-F based epoxy resin for specific applications. CAS # 9003-36-5 42423-25-6
	Phenol, 4,4'-methylenebis-, polymer with (chloromethyl)oxirane	42423-25-6				
ARFM-12 (XR-123)	A medium viscosity Bis-A/F blend	-	Max 2 (Gardner)	172 - 180	6,500 - 8,500	A medium viscosity Bis-A/F blend recommended for coatings, composites, construction applications and floor coatings.
ARFM-13 (XR-60)	A low viscosity Bis-A/F blend	-	Max 2 (Gardner)	174 - 182	4,500 - 6,500	A low viscosity Bis-A/F blend recommended for coatings, composites, construction applications and floor coatings.
ARFM-14 (XR-106)	A reactive diluent modified Bis-A/F blend	-	Max 2 (Gardner)	185 - 196	860 - 960	A reactive diluent modified Bis-A/F blend recommended for high solids coatings, construction and floor coatings.

EPOXY PHENOL NOVOLAC RESINS

Epoxy phenol novolac (EPN) resins are available in varying functionalities and are recommended to achieve higher chemical and thermal resistance in various applications including adhesives, coatings, composites, electrical and flooring.

Lapox® grade	Active substance	CAS No.	Colour / APHA	EEW g/eq	Viscosity @ 25°C mPa·s	Non-volatile content %	Recommendations
ARPN-25	A low viscosity EPN resin	-	Max 250	172 - 179	1,100 - 1,700 @ 52°C	-	A low viscosity EPN resin with average 2.5 functionality recommended for composites, electrical and coating applications.
ARPN-36 (L-238)	A standard semi-solid EPN resin	-	Max 250	175 - 182	20,000 - 50,000 @ 52°C	-	A standard semi-solid EPN resin having average 3.6 functionality recommended for composites, electrical, chemical resistant coatings and flooring.
ARPN-36 M 80	EPN resin	-	Max 23 (Gardner)	215 - 231	150 - 350	79-81	A solution of EPN resin ARPN-36 in MEK recommended for chemical resistant coatings, electrical and composite applications.
ARPN-36 X 80	EPN resin	-	Max 23 (Gardner)	215 - 231	800 - 1,500	79-81 150°C/1h)	A solution of EPN resin ARPN-36 in xylene recommended for chemical resistant coatings, electrical and composite applications.
ARPN-53 (XR-55)	A modified EPN resin with average 2.2 functionality	-	Max 43 (Gardner)	167 - 179	35,000 - 55,000		A modified EPN resin with average 2.2 functionality recommended for composites, electrical and coating applications.
ARPN-54	A modified EPN resin with average 2.2 functionality	-	Max 33 (Gardner)	167 - 182	25,000 - 35,000		

GLYCIDYL AMINE BASED MULTIFUNCTIONAL RESINS

Atul offers various specialty resins with different functionalities and thermal stabilities. Specialty resins provide high Tg (glass transition temperature) along with mechanical and thermal resistance, and are extensively used for electrical and electronic, FRP composites, high performance structural adhesives and prepregs.

Lapox® grade	Active substance	CAS No.	Colour / Gardner	EEW g/eq	Viscosity @ 25°C mPa·s	HyCl %	Recommendations
ARTF-13	Tetraglycidyl 4,4'-methylenedianiline (TGMDA)	28768-32-3	Max 12	117 - 134	7,000 - 11,000 @ 50°C	Max 0.10	Medium viscosity variants of tetrafunctional resin based on MDA for aerospace and high performance composites. CAS # 28768-32-3
ARTF-14	Tetraglycidyl 4,4'-methylenedianiline (TGMDA)	28768-32-3	Max 12	117 - 134	10,000 - 12,000 @ 50°C	Max 0.10	
ARTF-15	Tetraglycidyl 4,4'-methylenedianiline (TGMDA)	28768-32-3	Max 12	117 - 134	11,000 - 13,000 @ 50°C	Max 0.10	
ARTF-16	Tetraglycidyl 4,4'-methylenedianiline (TGMDA)	28768-32-3	Max 12	117 - 134	13,000 - 15,000 @ 50°C	Max 0.10	
ARTF-17	Tetraglycidyl 4,4'-methylenedianiline (TGMDA)	28768-32-3	Max 12	117 - 134	17,000 - 19,000 @ 50°C	Max 0.10	A high viscosity, tetrafunctional resin based on MDA for aerospace and high performance composites. CAS # 28768-32-3
ARTF-18	Tetraglycidyl 4,4'-methylenedianiline (TGMDA)	28768-32-3	Max 12	117 - 134	7,000 - 19,000 @ 50°C	Max 0.10	A general purpose tetrafunctional resin based on MDA for aerospace and high performance composites. CAS # 28768-32-3
ARTF-23 (XR-23)	Tetraglycidyl 4,4'-methylenedianiline (TGMDA)	28768-32-3	Max 12	111 - 117	3,000 - 6,000 @ 50°C	Max 0.10	A low viscosity, tetrafunctional resin based on MDA for aerospace and high performance composites. CAS # 28768-32-3
ARTF-33 (XR-93)	N,N,N',N'-tetraglycidyl-4,4'-diamino-3,3'-diethyldiphenylmethane	130728-76-6	Max 7	118 - 133	7,000 - 12,000	Max 0.10	A tetrafunctional resin based on ethyl substituted MDA for aerospace and high performance applications. The product offers very low viscosity and reactivity. CAS # 130728-76-6
ARTF-35	4-(Diglycidylamino)phenyl glycidyl ether	5026-74-4	–	105 - 115	2,000 - 5,000	Max 0.30	A trifunctional unmodified resin based on p-amino phenol for aerospace and high performance applications. CAS # 5026-74-4
ARTF-36	4-(Diglycidylamino)phenyl glycidyl ether	5026-74-4	–	95 - 106	550 - 850	Max 0.20	A distilled trifunctional unmodified resin based on p-amino phenol for aerospace and high performance applications. CAS # 5026-74-4
ARTF-37	m-(2,3-Epoxypropoxy)-N,N-bis(2,3-epoxypropyl)aniline	71604-74-5	–	102 - 110	7,000 - 13,000	Max 0.30	A trifunctional unmodified resin based on m-amino phenol for aerospace and high performance applications. CAS # 71604-74-5
ARTF-38	m-(2,3-Epoxypropoxy)-N,N-bis(2,3-epoxypropyl)aniline	71604-74-5	–	94 - 102	1,500 - 4,800	Max 0.20	A distilled trifunctional unmodified resin based on m-amino phenol for aerospace and high performance applications. CAS # 71604-74-5
ARTF-32	4-[[4-[bis(oxiran-2-ylmethyl)amino]-3-methylphenyl]methyl]-2-methyl-N,N-bis(oxiran-2-ylmethyl)aniline	142675-09-0	Max 18	118 - 135	2,000 - 4,000 @ 50°C	Max 0.10	A low viscosity, tetrafunctional resin based on substituted MDA for aerospace and high performance composites. CAS # 142675-09-0
ARTF-39	A modified low viscosity, multifunctional resin	–	Max 10	125 - 143	2,500 - 4,000 @ 50°C	Max 0.10	A modified low viscosity, multifunctional resin for aerospace and high performance composites.

03 / 03

Hardeners

Technical data of curing agents

Hardeners

Technical data of curing agents for various curing regimes.

Active substance	CAS No.	Appearance	Melting point °C	Purity by HPLC %	Ash content %	Median particle size μ
4,4'-Diaminodiphenyl sulfone	80-08-0	White to off-white powder	176 – 185	Min 99.0	Max 0.10	–
Micronised 4,4'-Diaminodiphenyl sulfone	80-08-0	White to off-white powder	176 – 180	Min 99.0	Max 0.10	5 – 15
Free-flowing 4,4'-Diaminodiphenyl sulfone	80-08-0	White to off-white powder	176 – 185	Min 99.0	Max 0.10	–
3,3'-Diaminodiphenyl sulfone	599-61-1	White to brownish white powder	167 – 175	Min 99.0	Max 0.10	–
Micronised 3,3'-Diaminodiphenyl sulfone	599-61-1	Off-white to yellowish brown powder	167 – 175	Min 99.0	Max 0.10	Max 10